

Class 08: Breast Cancer Mini Project

Mankeerat Rataul

Table of contents

Background	1
Data Import	1
Exploratory Data Analysis	3
Performing PCA	4
Interpreting PCA Results	5
Variance explained	8
Communicating PCA results	10
Hierarchical Clustering	11
Combining methods	14
Prediction	17

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a data-set from fine needle aspiration (FNA) of breast mass.

Data Import

read.csv is used here to import the csv file into a dataframe `wisc.df`

```
fna.data <- "WisconsinCancer.csv"
wisc.df <- read.csv(fna.data, row.names=1)
head(wisc.df, 3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.8	1001
842517	M	20.57	17.77	132.9	1326
84300903	M	19.69	21.25	130.0	1203

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.6	2019	0.1622	0.6656
842517	158.8	1956	0.1238	0.1866
84300903	152.5	1709	0.1444	0.4245

	concavity_worst	concave.points_worst	symmetry_worst
842302	0.7119	0.2654	0.4601
842517	0.2416	0.1860	0.2750
84300903	0.4504	0.2430	0.3613

	fractal_dimension_worst
842302	0.11890
842517	0.08902
84300903	0.08758

Make sure we remove or exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[,-1]
head(wisc.data, 3)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
842302	17.99	10.38	122.8	1001	0.11840
842517	20.57	17.77	132.9	1326	0.08474

84300903	19.69	21.25	130.0	1203	0.10960
	compactness_mean	concavity_mean	concave.points_mean	symmetry_mean	
842302	0.27760	0.3001		0.14710	0.2419
842517	0.07864	0.0869		0.07017	0.1812
84300903	0.15990	0.1974		0.12790	0.2069
	fractal_dimension_mean	radius_se	texture_se	perimeter_se	area_se
842302	0.07871	1.0950	0.9053	8.589	153.40
842517	0.05667	0.5435	0.7339	3.398	74.08
84300903	0.05999	0.7456	0.7869	4.585	94.03
	smoothness_se	compactness_se	concavity_se	concave.points_se	
842302	0.006399	0.04904	0.05373	0.01587	
842517	0.005225	0.01308	0.01860	0.01340	
84300903	0.006150	0.04006	0.03832	0.02058	
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.6	2019	0.1622	0.6656	
842517	158.8	1956	0.1238	0.1866	
84300903	152.5	1709	0.1444	0.4245	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				

```
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

There are 569 individual patients.

Q2. How many of the observations have a malignant diagnosis?

```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

212 individuals have a malignant diagnosis.

Q3. How many variables/features in the data are suffixed with `_mean`?

```
length(grep("_mean", colnames(wisc.data)))
```

```
[1] 10
```

10 total variables suffixed with `_mean`.

Performing PCA

We need to scale our data before PCA w/ the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale =T)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987

Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427 of the original variance is captured.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

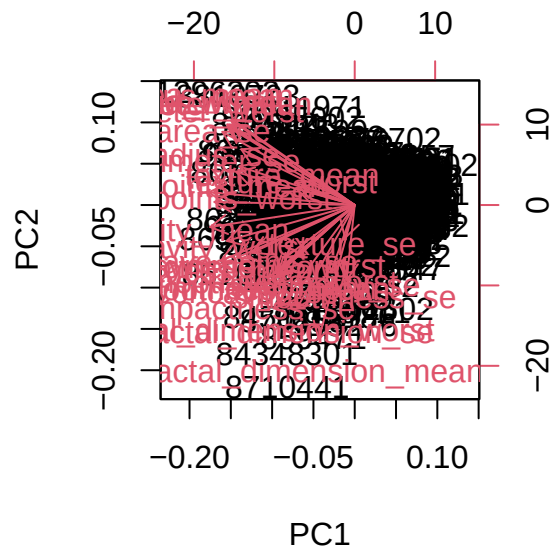
The first 3 PCs are needed to describe at minimum 70% of the variance.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 Principal components are required to describe >90% of the original variance.

Interpreting PCA Results

```
biplot(wisc.pr)
```



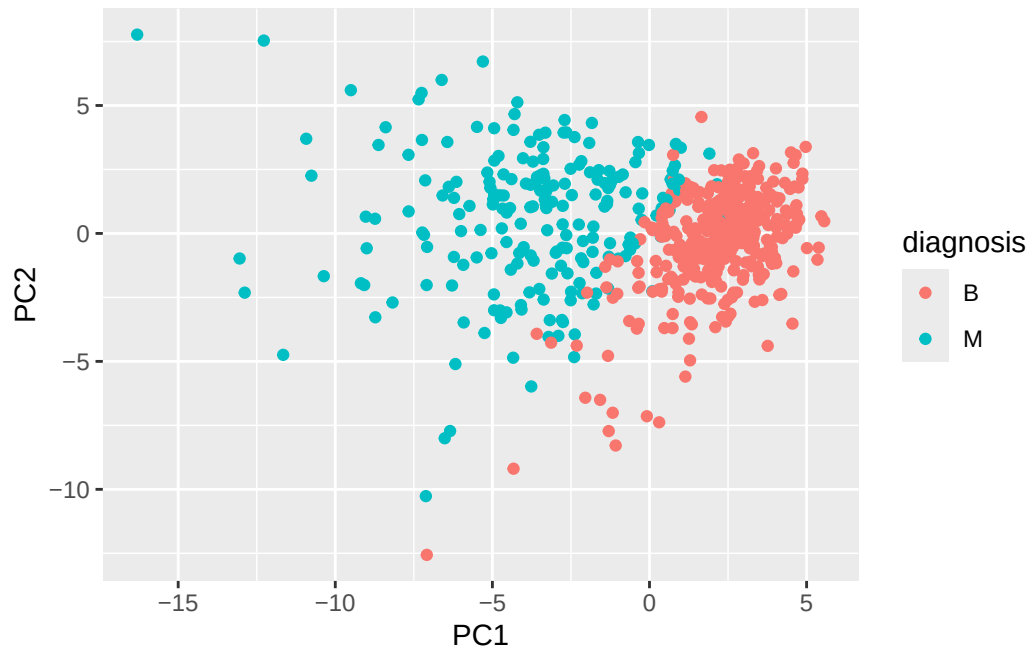
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot has an incredible amount of data that cannot be interpreted. It shows the directions in which each variable pulls the Principle Components, alongside listing all 569 text entries for individual patients overlapping. It is extremely difficult to understand for that very reason, since you cannot read any of the data.

The following is the use of a ggplot with diagnostic data graphed over.

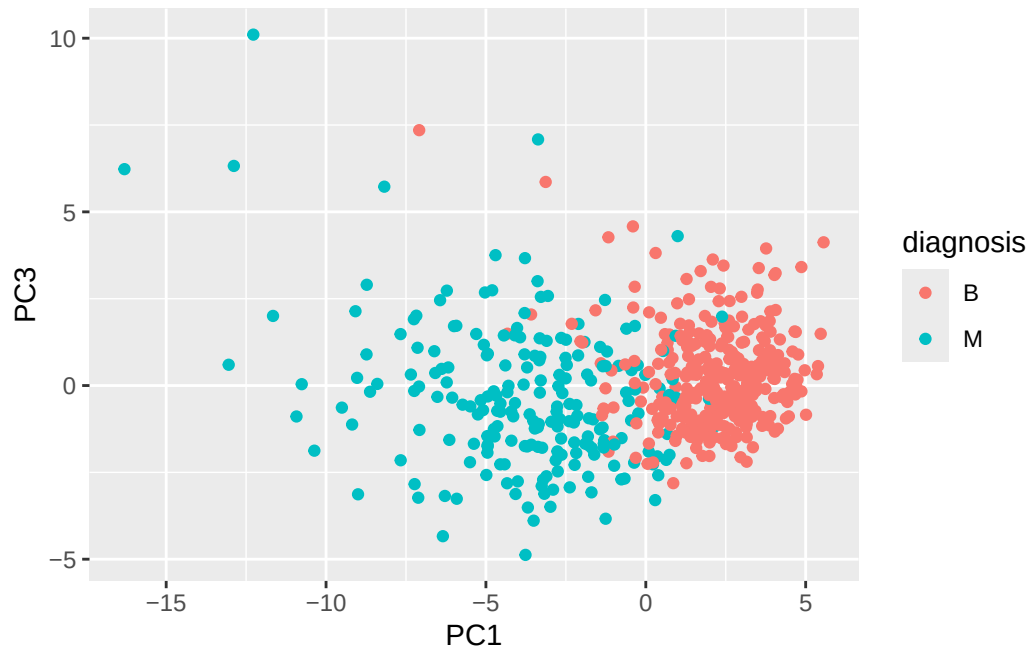
```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



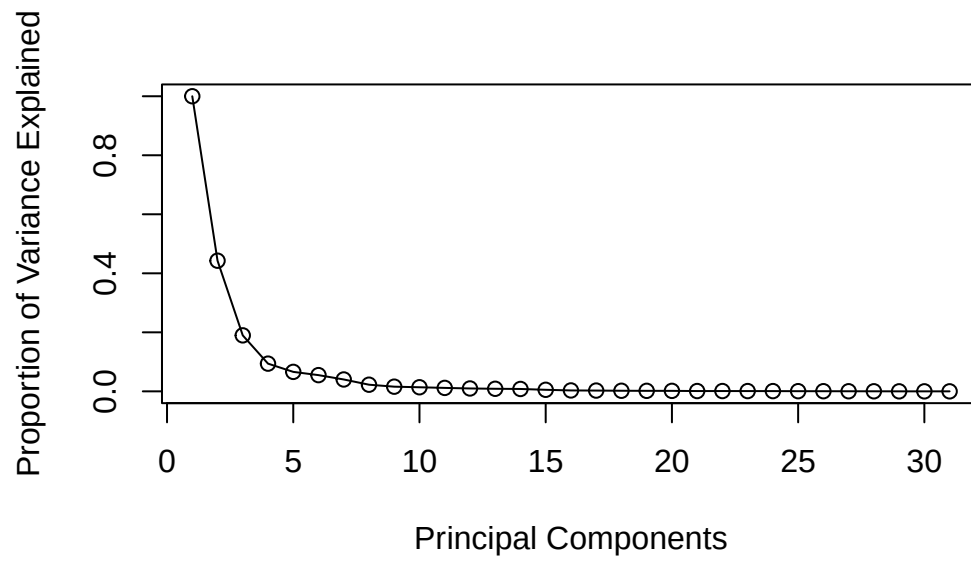
There seems to be a clear clustering of benign tumors vs malignant with respect to these principle components.

Variance explained

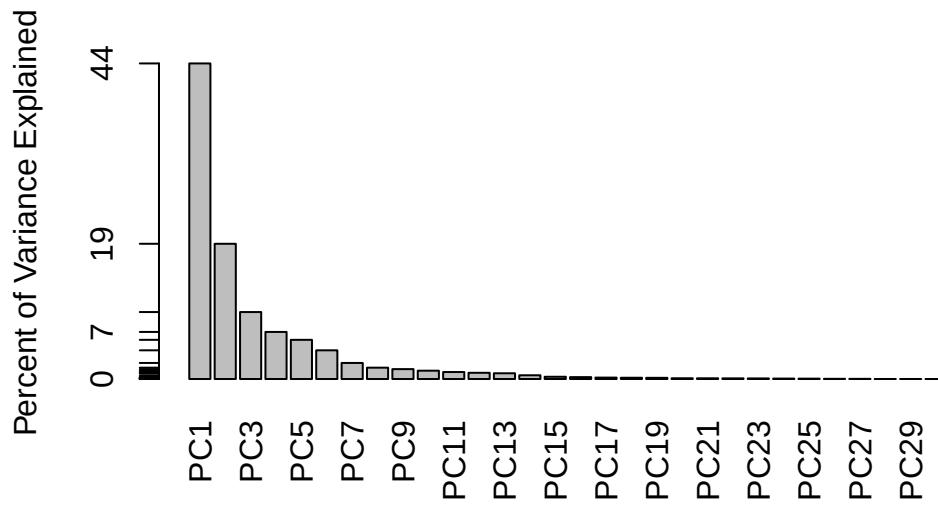
```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var/(sum(pr.var))
plot(c(1, pve), xlab = "Principal Components", ylab="Proportion of Variance Explained", ylim=
```

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
variancofpr1 <- wisc.pr$rotation[,1]
variancofpr1["concave.points_mean"]
```

```
concave.points_mean
-0.2608538
```

```
variancofpr1
```

radius_mean	texture_mean	perimeter_mean
-0.21890244	-0.10372458	-0.22753729
area_mean	smoothness_mean	compactness_mean
-0.22099499	-0.14258969	-0.23928535
concavity_mean	concave.points_mean	symmetry_mean
-0.25840048	-0.26085376	-0.13816696

fractal_dimension_mean	radius_se	texture_se
-0.06436335	-0.20597878	-0.01742803
perimeter_se	area_se	smoothness_se
-0.21132592	-0.20286964	-0.01453145
compactness_se	concavity_se	concave.points_se
-0.17039345	-0.15358979	-0.18341740
symmetry_se	fractal_dimension_se	radius_worst
-0.04249842	-0.10256832	-0.22799663
texture_worst	perimeter_worst	area_worst
-0.10446933	-0.23663968	-0.22487053
smoothness_worst	compactness_worst	concavity_worst
-0.12795256	-0.21009588	-0.22876753
concave.points_worst	symmetry_worst	fractal_dimension_worst
-0.25088597	-0.12290456	-0.13178394

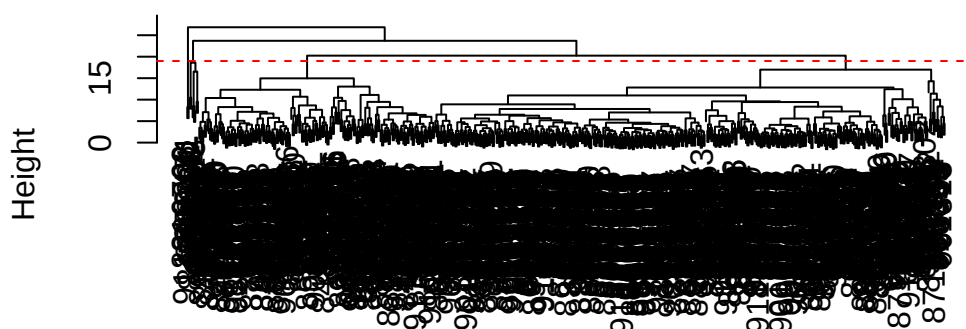
It seems to have been around -0.261. This means that this feature contributed 26.1% against PC1. There seems to be no other features with larger contributions than this one.

Hierarchical Clustering

Clustering and plotting is done below:

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```

Cluster Dendrogram



```
dist(scale(wisc.data))
hclust(*, "complete")
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

19

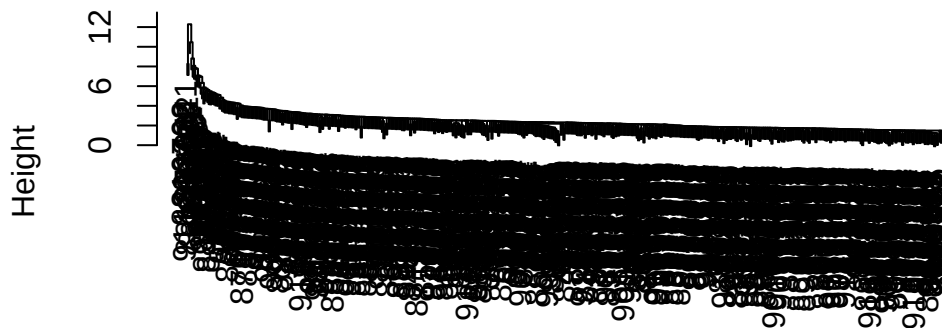
```
wisc.hclust.clusters <- cutree(wisc.hclust, h=19)
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

```
wistsingle.hclust <- hclust(dist(scale(wisc.data)), method = "single")
wistavg.hclust <- hclust(dist(scale(wisc.data)), method = "average")
wistwardD2.hclust <- hclust(dist(scale(wisc.data)), method = "ward.D2")
plot(wistsingle.hclust, main="Single method")
```

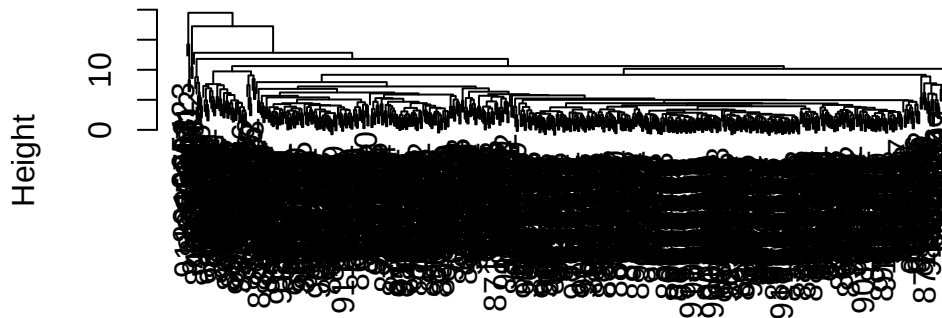
Single method



```
dist(scale(wisc.data))  
hclust (*, "single")
```

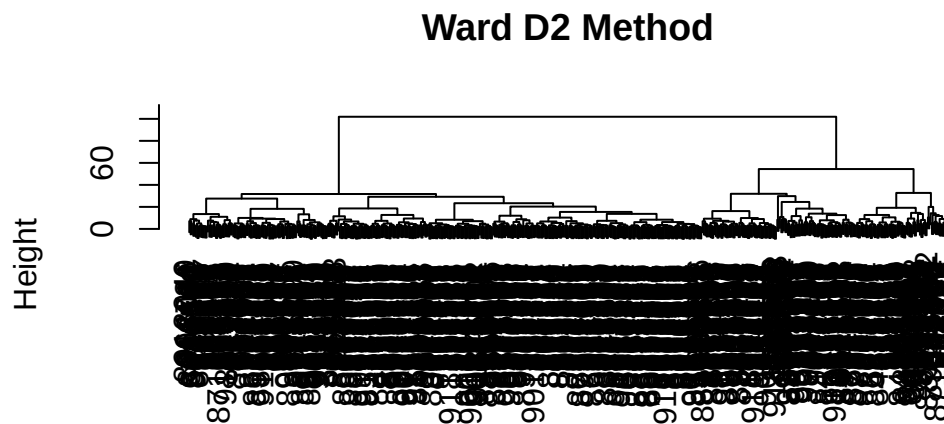
```
plot(wistavg.hclust, main="Average Method")
```

Average Method



```
dist(scale(wisc.data))  
hclust (*, "average")
```

```
plot(wistwardD2.hclust, main="Ward D2 Method")
```



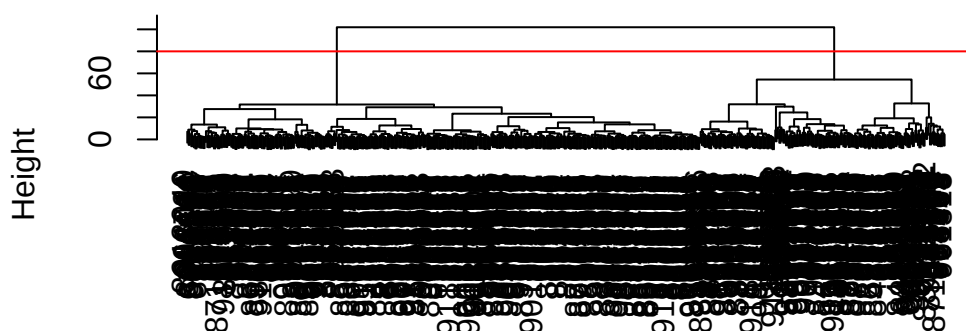
```
dist(scale(wisc.data))
hclust (*, "ward.D2")
```

Ward D2 is my favorite results for this, as it gives 2 clear groups and would seem to work best at splitting into 2 separate groups.

Combining methods

```
d <- dist(wisc.pr$x[,1:30])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=80, col="red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

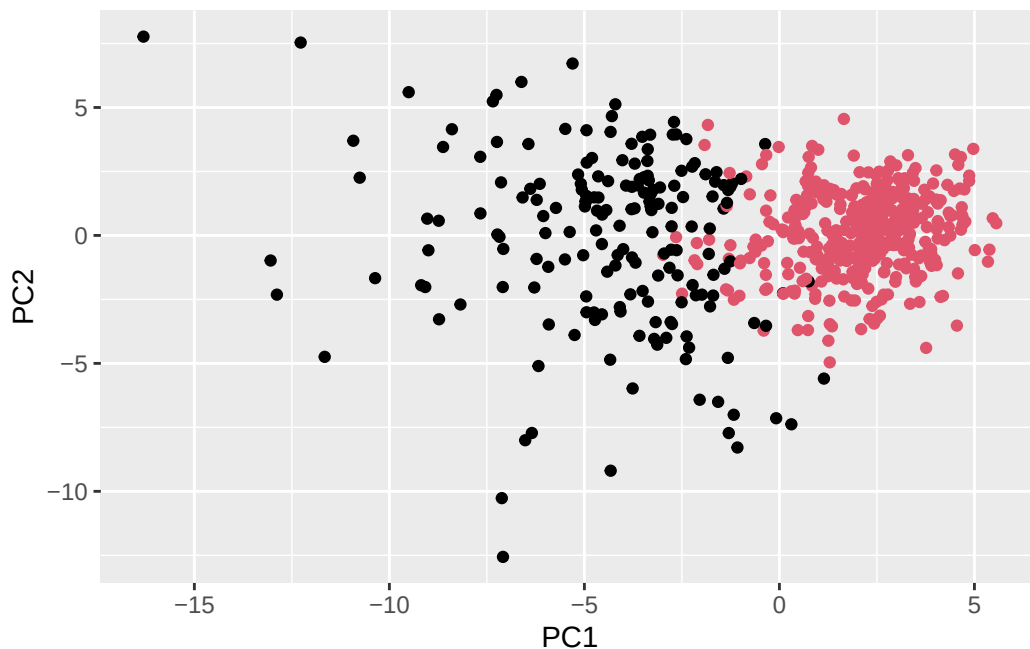
```
grps
  1  2
184 385
```

How does this clustering `grps` correspond to the expert diagnosis?

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
B      20 337
M     164  48
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

It seems to separate it out very well, with one group containing 164 true malignancies and 20 false positives, and the other group containing 337 true benign cases and 48 false negatives.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
	B	M
wisc.hclust.clusters		
1	12	165
2	2	5
3	343	40
4	0	2

It seems that it did fairly well. Separating out the outliers of cluster 2 and 4, it has a cluster containing 165 true malignancies and 12 false positives, and a cluster with 343 true benigns and 40 false positives. In a way it had around the same strength as the PCA analysis.

Prediction

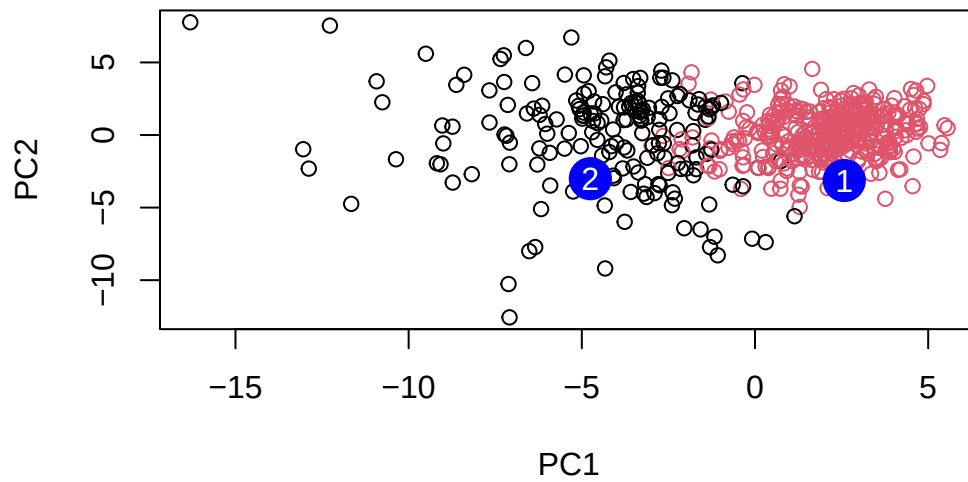
I will use the `predict()` function below to take the PCA model and predict the diagnosis for new cancer cell data by projection onto the PCA.

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

Above are the PCA outputs

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Most likely the patient at position 2, as this lines up directly with prior analysis of likelihood to have malignancy.